

PARTH SARTHI SAXENA

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PROFESSIONAL SUMMARY

Quantitative researcher and systematic trader with hands-on experience building and validating trading strategies across Indian equity and commodity futures markets. Developed 2 fully backtested strategies with real cost models, out-of-sample validation, and statistical significance testing. Executed 400+ live trades in NSE equities and F&O. Strong foundations in Python, C++, probability, statistics, and time-series analysis. Solved 300+ LeetCode problems in C++. Seeking quant research and algo trading internships.

EDUCATION

JECRC Foundation, Jaipur

Expected May 2027

B.Tech – Electronics & Communication Engineering | CGPA: 8.21/10 (~3.4/4.0)

Relevant Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Numerical Methods, Operating Systems

TRADING & QUANTITATIVE EXPERIENCE

Independent Systematic Trader & Quant Researcher

Jun 2024 – Present

NSE Equities & F&O — Live Capital

- Executed 400+ live trades in NSE equities and F&O using systematic rule-based strategies — maintaining documented win rates of 51–56%, risk-reward ratios of 1:3–1:5, and controlled drawdown of ~10–12%
- Developed and validated 2 fully backtested quantitative strategies across 10+ years of real market data — including realistic transaction cost models, slippage, STT, brokerage, and Indian progressive tax calculations
- Applied expected value framework, Z-score signal generation, and regime filtering to systematically identify and validate statistical edge before live deployment
- Conducted out-of-sample validation across 3 independent time splits — confirming strategy robustness and absence of overfitting (p-value: 0.011 vs random baseline)
- Maintained structured trade journal and performance attribution across all live positions — tracking Sharpe ratio, CAGR, drawdown, and win rate at portfolio level

QUANTITATIVE RESEARCH PROJECTS

GSR Alpha — Gold/Silver Ratio Mean-Reversion | MCX Futures

2025 – 2026

GitHub: github.com/parthsarthisaxena/GSR-Alpha

- Designed and backtested a statistical arbitrage strategy exploiting mean-reversion in the Gold/Silver Ratio on MCX Futures — 10 years of real market data (2,511 trading days, 76 trades)
- Achieved net return of +170.88% | Sharpe: 4.91 | Max Drawdown: -14.50% | Calmar: 11.78 | Win Rate: 63.2% | Profit Factor: 2.36 — starting capital ₹5,00,000 growing to ₹13,54,400
- Built full real-cost model including brokerage, STT, exchange charges, GST, SEBI charges, slippage, and Indian progressive income tax (marginal slab rates, not flat 30%) — total friction ~0.28–0.35% per round-trip
- Validated out-of-sample across 3 independent time splits (all profitable); rejected pairs-trading and Kelly criterion variants; confirmed statistical significance (p=0.011, no look-ahead bias, no autocorrelation inflation)
- Systematically tested and documented 3 prior failing strategies before arriving at this solution — demonstrating rigorous iterative research methodology

Nifty 50 EMA Reversion Strategy — Intraday Backtest

Apr 2025 – May 2025

GitHub: github.com/parthsarthisaxena/nifty-ema-reversion-strategy

- Built a complete intraday mean-reversion backtesting system on 20 Nifty 50 stocks over 1 year — simulating 564 trades with partial exit management (TP1 at 0.9%, TP2 at 2.5%) and regime filtering via 20-day EMA
- Modeled all real transaction costs — identifying that transaction cost drag (-23% net vs +90% gross) was the core strategy limitation, not signal quality; win rate of 46.6% confirmed genuine statistical edge above breakeven threshold
- Documented 4 iterative research versions from initial breakout hypothesis to regime-filtered EMA reversion — demonstrating systematic hypothesis formation, testing, and rejection methodology
- Identified 3 quantified fixes (futures migration, larger profit targets, volatility filtering) and validated signal quality via manual paper trades showing 5.74 risk-reward on individual setups

SKILLS

Programming: Python (pandas, numpy, scipy, matplotlib, yfinance), C++, SQL

Quantitative: Backtesting, Statistical Arbitrage, Mean-Reversion, Expected Value, Z-score Signals, Regime Filtering, OOS Validation, Performance Attribution, Options Pricing (Black-Scholes), Kelly Criterion

Markets: NSE Equities, F&O, MCX Futures, Derivatives, Transaction Cost Modeling, Market Microstructure, Indian Tax Structure

Statistics: Probability, Time-Series Analysis, Hypothesis Testing, Sharpe / Calmar / Sortino Ratios, Drawdown Analysis, Autocorrelation

CS Fundamentals: Data Structures & Algorithms, OOP, Git, Clean Code — 300+ LeetCode problems solved in C++